Spectral Poisson–Dirichlet Universality for Luce Permutations under Arbitrary Rate Arrangements

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Abstract

A Luce permutation is obtained by ranking independent exponential clocks with prescribed positive rates. We prove an arrangement-free criterion for its macroscopic cycles: if the empirical normalized rate law converges to a law whose exponential observation channel has a strict mean-zero L^2 spectral gap, then the size-biased cycle lengths converge in ℓ^1 to GEM(1) and the ranked cycle lengths converge in ℓ^1 to PD(1). Only the asymptotic rate spectrum matters; the rates may be assigned to the labels in an arbitrary deterministic order.

The proof identifies two mechanisms. A factorization through the marginal rate channel shows that label–rate coupling cannot worsen its spectral gap, and an exact finite-state Markov contraction controls all late clock rows by their total mass, without a Hilbert–Schmidt assumption. A dyadic Poincaré inequality combined with Karamata theory establishes the channel gap for every rate law with a regularly varying lower tail of positive index. Consequently the theorem covers all uniformly comparable arrays, every rearrangement of the Sukhatme rates $1, 2, \dots, n$, every rearrangement of $1^\gamma, 2^\gamma, \dots, n^\gamma$ for $\gamma > 0$, and general positive regularly varying spectra. In particular, it resolves the macroscopic- and longest-cycle problem for the Sukhatme model. We also give a subsequential criterion and a quenched extension to random rate environments. Two further spectral results identify a broader sufficient condition and a boundary of the criterion: successive dyadic shell-mass ratios bounded away from zero and one imply a strict channel gap, whereas a slowly varying lower tail has channel norm one.

Keywords. Luce permutations; Poisson–Dirichlet distributions; exponential races; spectral gaps; regular variation.

1 Introduction

Let $X_{n,1}, \dots, X_{n,n}$ be independent exponential random variables with positive rates $\theta_{n,1}, \dots, \theta_{n,n}$. Their increasing order defines the label-to-position permutation

$$\sigma_n(i) = 1 + \#\{j : X_{n,j} < X_{n,i}\}. \tag{1}$$

This is the exponential-race representation of the Plackett–Luce ranking model; see Luce [13], Plackett [15], and Chatterjee–Diaconis–Kim [4]. Multiplying all rates by one positive constant leaves the permutation unchanged.

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The cycle structure of a uniform random permutation is governed by the Ewens parameter-one law [7]. Its normalized ranked cycles converge to Kingman's PD(1) distribution, while their size-biased order converges to the GEM(1) residual-allocation law; see Kingman [10] and Pitman [14, Chapter 3]. For Luce permutations the entry law, permuton, local patterns, and short-cycle behavior may be highly nonuniform. The question addressed here is whether the macroscopic cycle partition nevertheless retains the uniform-permutation limit.

Recent Luce models and the cycle questions

Chatterjee, Diaconis, and Kim [4] developed the enumerative theory of the Tsetlin library and explicitly asked for the number of cycles, the longest cycle, the order, and the longest increasing subsequence of a Luce-distributed permutation. They also singled out the decreasing weights $n, n-1, \dots, 1$, arising from exponential spacings, and called them the Sukhatme weights. Borga, Chatterjee, and Diaconis [1] subsequently proved permuton and local-limit theorems, limiting pattern laws, a quenched Benjamini–Schramm limit, and an inversion central limit theorem. Their Sukhatme permuton is singular at an endpoint and its pattern statistics are far from uniform, making the macroscopic-cycle question especially natural.

A related but different source of Poisson–Dirichlet limits is the model of random standardized permutations studied by Guerder [8]. In that model one standardizes an i.i.d. word, and a uniform upper bound on the largest letter probability yields a PD(1) limit for large cycles. The present model instead starts from independent continuous clocks with a deterministic triangular array of rates. Our criterion is spectral, permits rates arbitrarily close to zero, and is invariant under every deterministic assignment of a prescribed asymptotic rate spectrum to the labels.

This arrangement invariance is a central point. Permuton and local-statistic limits naturally retain the coupling between labels and rates [1]; the macroscopic cycle partition proved below depends only on the limiting marginal distribution of the normalized rates.

This paper extends the author's earlier comparable-rate theorem [11], whose full scope, including arrays without a limiting empirical rate law, is recovered in Corollary 8.1. We retain its clock-slab and residual permanental framework, adaptive relative-prefix estimates, rooted exploration, and deletion-regeneration argument, reproducing the necessary proofs here. The new ingredients are domination by the rate-observation channel, a late-row contraction estimate that avoids Hilbert–Schmidt assumptions, and spectral-gap criteria for endpoint-singular rate laws. These extend the macroscopic cycle limits to Sukhatme and positive-index regularly varying spectra under arbitrary deterministic arrangements.

The exponential channel

Normalize

$$r_{n,i} = \frac{\theta_{n,i}}{\max_j \theta_{n,j}} \in (0,1]$$

and let

$$\nu_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{r_{n,i}} \tag{2}$$

be the empirical normalized rate law. Suppose first that $\nu_n \Rightarrow \nu$ on $[0,1]$. The limiting exponential channel is obtained by sampling $R \sim \nu$ and then sampling $T \mid R = r \sim \mathrm{Exp}(r)$. Its time marginal has density

$$B_\nu(t) = \int_{(0,1]} r e^{-rt}\, \nu(\,dr), \qquad \mu_\nu(\,dt) = B_\nu(t)\,dt. \tag{3}$$

When $\nu(\{0\}) = 0$, μ_ν is a probability measure. Define

$$(C_\nu f)(r) = \int_0^\infty f(t) r e^{-rt}\, dt, \qquad C_\nu : L^2(\mu_\nu) \longrightarrow L^2(\nu). \tag{4}$$

This is conditional expectation of $f(T)$ given $R = r$. It preserves constants and is an L^2 contraction. Its mean-zero norm

$$q(\nu) = \|C_\nu\|_{L^2_0(\mu_\nu)\to L^2_0(\nu)} \tag{5}$$

is the maximal L^2 correlation between the latent rate and its observed clock time. It is the only global mixing parameter needed below.

Let $U_1, U_2, \dots$ be independent Unif$(0,1)$ variables. We write GEM(1) for

$$U_1, \quad (1-U_1)U_2, \quad (1-U_1)(1-U_2)U_3, \dots,$$

and PD(1) for its decreasing rearrangement. Put

$$\mathcal{S}^\downarrow = \left\{ x_1 \ge x_2 \ge \cdots \ge 0 : \sum_{j\ge 1} x_j = 1 \right\},$$

with the ℓ^1 metric. If $L_{n,1} \ge L_{n,2} \ge \cdots$ are the cycle lengths of σ_n, let $\widetilde{L}_{n,1}, \widetilde{L}_{n,2}, \dots$ be their size-biased deletion order: at each stage choose a vertex uniformly from the residual set, record its cycle length, delete that cycle, and repeat.

Theorem 1.1 (Spectral Poisson–Dirichlet criterion)**.** *Assume that*

$$\nu_n \Rightarrow \nu, \qquad \nu(\{0\}) = 0, \qquad q(\nu) < 1.$$

Then

$$\left(\frac{\widetilde{L}_{n,1}}{n}, \frac{\widetilde{L}_{n,2}}{n}, \dots \right) \Rightarrow \mathrm{GEM}(1) \quad \text{in } \ell^1, \tag{6}$$

and

$$\left(\frac{L_{n,1}}{n}, \frac{L_{n,2}}{n}, \dots \right) \Rightarrow \mathrm{PD}(1) \quad \text{in } \mathcal{S}^\downarrow. \tag{7}$$

If I_n is a uniformly chosen label and $\mathscr{L}_n(I_n)$ is the length of its cycle, then

$$\frac{\mathscr{L}_n(I_n)}{n} \Rightarrow \mathrm{Unif}(0,1). \tag{8}$$

The conclusion depends only on the limiting rate marginal ν; no regularity assumption is imposed on the assignment of rates to labels.

The convergence of the empirical spectrum can be weakened to a condition on all of its limit points.

Corollary 1.2 (Subsequential spectral criterion)**.** *Suppose that every weak subsequential limit ν of (ν_n) satisfies $\nu(\{0\}) = 0$ and $q(\nu) < 1$. Then (6)–(8) hold.*

Proof. Every subsequence has a further subsequence along which ν_n converges. Theorem 1.1 identifies the same universal limit on every such refinement. The subsequence principle gives the result. □

The criterion also has a quenched random-environment form.

Corollary 1.3 (Random rate environments)**.** *Let the positive rate vectors $\boldsymbol{\theta}_n = (\theta_{n,1}, \dots, \theta_{n,n})$ be random and independent of the exponential variables used to generate the ranking. Suppose their empirical normalized laws ν_n converge in probability, in the weak topology, to a deterministic law ν satisfying $\nu(\{0\}) = 0$ and $q(\nu) < 1$. Conditional on $\boldsymbol{\theta}_n$, the laws in (6) and (7) converge in probability, in bounded-Lipschitz distance, to* GEM(1) *and* PD(1)*, respectively. In particular, the corresponding annealed convergences hold.*

Proof. From every subsequence one may extract a further subsequence along which $\nu_n \Rightarrow \nu$ almost surely. For almost every realization of the rate environment, Theorem 1.1 applies to the resulting deterministic triangular array, so the conditional bounded-Lipschitz distance tends to zero almost surely. The subsequence criterion for convergence in probability proves the quenched assertion, and bounded convergence gives the annealed one. □

The theorem gives the longest-cycle law explicitly.

Corollary 1.4 (Longest cycle and moment convergence)**.** *Under the hypotheses of Theorem 1.1, let P_1 denote the largest part of a* PD(1) *partition. Then*

$$\frac{L_{n,1}}{n} \Rightarrow P_1, \qquad \mathbb{E}\left(\frac{L_{n,1}}{n}\right)^s \longrightarrow \mathbb{E}P_1^s \quad (s > 0). \tag{9}$$

In particular,

$$\frac{\mathbb{E}L_{n,1}}{n} \longrightarrow \lambda_{\mathrm{GD}},$$

where $\lambda_{\mathrm{GD}} = \mathbb{E}P_1$ is the Golomb–Dickman constant. These conclusions answer the longest-cycle question of Chatterjee–Diaconis–Kim [4] for the Sukhatme weights, which they singled out as a natural example, and for every deterministic rearrangement of $1, \dots, n$.

Proof. The largest-coordinate map is continuous on $\mathcal{S}^{\downarrow}$ in the ℓ^1 metric. Equation (7) therefore gives the distributional limit. Since $0 \le L_{n,1}/n \le 1$, every positive moment converges. The identification of $\mathbb{E}P_1$ with the Golomb–Dickman constant is classical; see Shepp and Lloyd [17]. □

The theorem includes the uniformly comparable regime: support in $[c, 1]$ gives a common exponential component and hence $q(\nu) \le \sqrt{1-c} < 1$. The new phenomenon is that support may reach zero. Our principal sufficient condition is regular variation.

Theorem 1.5 (Regularly varying channel gap)**.** *Let ν be a probability law on $(0, 1]$, and put*

$$F_\nu(x) = \nu((0, x]).$$

If F_ν is regularly varying at zero with index $\alpha > 0$, then $q(\nu) < 1$. In particular, for every $\alpha > 0$,

$$\nu_\alpha(\,dr) = \alpha r^{\alpha-1}\mathbf{1}_{(0,1)}(r)\,dr \tag{10}$$

has a strict channel gap.

Corollary 1.6 (Power-law rate spectra)**.** *Fix $\gamma > 0$. Suppose that, for each n, the multiset of rates is*

$$\{1^\gamma, 2^\gamma, \dots, n^\gamma\},$$

assigned to the labels in an arbitrary deterministic order. Then (6)–(8) and (9) hold. The case $\gamma = 1$ includes every rearrangement of the Sukhatme rates.

Corollary 1.7 (Regularly varying spectra)**.** *Let $(a_m)_{m\ge1}$ be a positive regularly varying sequence of index $\gamma > 0$, and for each n assign the multiset $\{a_1, \dots, a_n\}$ to the labels in an arbitrary deterministic order. Then (6)–(8) and (9) hold.*

Regular variation is used to identify the limiting empirical spectrum and to prove its spectral gap. We use the uniform convergence theorem, Potter bounds, and Karamata's Abelian theorem from Bingham–Goldie–Teugels [2]. Standard compactness, maximal inequalities, and weak-convergence facts are taken from Kallenberg [9], Durrett [6], Billingsley [3], and Dudley [5].

Two structural points drive the proof. First, a marked subsequential limit remembers the arrangement of rates along the labels, but its channel factors through C_ν and therefore cannot have a worse mean-zero norm. Second, late clock rows are controlled by the exact finite-state inequality

$$\sum_b \frac{(\sum_{a\in D} p_a P_{ab})^2}{p_b} \le p(D)$$

for every stochastic matrix P with positive stationary law p. This replaces the global Hilbert–Schmidt estimate that would fail for the Sukhatme kernel. Sections 2–6 develop the spectral and regeneration argument, Section 7 proves the regularly varying gap, Section 8 gives the principal applications, and Section 9 proves the dyadic-doubling criterion and the slowly varying boundary theorem before recording the remaining spectral and multiscale questions.

2 The limiting exponential channel

2.1 Marked subsequential limits

Fix an arbitrary subsequence. Since $[0,1]^2$ is compact, after passage to a further subsequence the marked empirical measures

$$\eta_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{(i/n, r_{n,i})} \tag{11}$$

converge weakly to a probability measure η on $[0,1]^2$. Its first marginal is Lebesgue measure and its second marginal is ν. Choose a disintegration

$$\eta(\,dx,\,dr) = \,dx\,\eta_x(\,dr). \tag{12}$$

Since $\nu(\{0\}) = 0$, we have $\eta_x(\{0\}) = 0$ for almost every x.

Put

$$A(t) = \int_{(0,1]} (1 - e^{-rt})\nu(\,dr), \qquad B(t) = A'(t) = B_\nu(t). \tag{13}$$

Then A is continuous and strictly increasing from 0 to 1. Let $\tau = A^{-1} : (0,1) \to (0,\infty)$ and define

$$g_x(t) = \int_{(0,1]} re^{-rt}\eta_x(\,dr), \qquad \rho(x,y) = \frac{g_x(\tau(y))}{B(\tau(y))}. \tag{14}$$

Lemma 2.1 (Bistochastic marked kernel)**.** *The kernel ρ is bistochastic up to null sets:*

$$\int_0^1 \rho(x,y)\,dy = 1 \quad \text{for a.e. } x, \qquad \int_0^1 \rho(x,y)\,dx = 1 \quad \text{for a.e. } y.$$

Proof. Under the change of variables $y = A(t),\ dy = B(t)\,dt$, and therefore

$$\int_0^1 \rho(x,y)\,dy = \int_0^\infty g_x(t)\,dt = \int_{(0,1]} \eta_x(\,dr) = 1.$$

Also $\int_0^1 g_x(t)\,dx = B(t)$, which proves the column identity. □

Let K_η be the integral operator with kernel ρ on $L^2[0,1]$. The next factorization explains why the arrangement of rates cannot worsen the spectral contraction of their marginal law.

Lemma 2.2 (Channel domination)**.** *On mean-zero subspaces,*

$$\|K_\eta\|_{L^2_0 \to L^2_0} \le q(\nu). \tag{15}$$

Proof. Let $U : L^2(0,1;\,dy) \to L^2(\mu_\nu)$ be the unitary map $(Uf)(t) = f(A(t))$. Define

$$(C_\eta h)(x) = \int_0^\infty h(t) g_x(t)\,dt$$

after which $K_\eta = C_\eta U$. For $k \in L^2(\nu)$ put

$$(Jk)(x) = \int k(r)\eta_x(\,dr).$$

Jensen's inequality gives

$$\|Jk\|^2_{L^2(\,dx)} \le \int_0^1 \int |k(r)|^2 \eta_x(\,dr)\,dx = \|k\|^2_{L^2(\nu)}.$$

Fubini's theorem shows that $C_\eta = JC_\nu$. Both factors preserve constants and means. Hence the mean-zero norm of C_η is at most $q(\nu)$, and the unitary map U does not change the norm. □

Remark 2.3. The quantity $q(\nu)$ is the maximal L^2 correlation between the latent rate R and its exponential observation T. We use only the operator definition (5); no compactness of the channel is assumed.

3 Clock slabs and residual permanental matchings

Work along the marked subsequence fixed in Section 2. Replace the original clocks by

$$T_{n,i} = (\max_j \theta_{n,j}) X_{n,i} \sim \mathrm{Exp}(r_{n,i});$$

their order, and hence σ_n, is unchanged.

Fix $T > 0$ and a finite partition

$$0 = t_0 < t_1 < \cdots < t_d = T, \qquad \max_a (t_a - t_{a-1}) \le h, \tag{16}$$

and append $I_\star = (T, \infty)$. Put

$$I_a = (t_{a-1}, t_a], \qquad H_{n,a} = \{i : T_{n,i} \in I_a\}, \qquad m_{n,a} = |H_{n,a}|.$$

The tail state is defined analogously. If $M_{n,a} = \sum_{c \le a} m_{n,c}$, the ranks occupied by $H_{n,a}$ form the interval

$$Q_{n,a} = \{M_{n,a-1} + 1, \dots, M_{n,a}\}.$$

Define

$$N_{n,ab} = |Q_{n,a} \cap H_{n,b}|, \qquad p_{n,a} = \frac{m_{n,a}}{n}, \qquad P_{n,ab} = \frac{N_{n,ab}}{m_{n,a}}. \tag{17}$$

On the high-probability events used below all rows are nonempty. The matrix P_n is stochastic and $p_n P_n = p_n$.

Let

$$J_a = (A(t_{a-1}), A(t_a)], \qquad J_\star = (A(T), 1], \qquad p_a = |J_a|,$$

and

$$q_{ab} = \int_{J_a \times J_b} \rho(x, y)\, dx\, dy, \qquad P_{ab} = q_{ab}/p_a. \tag{18}$$

Every p_a and q_{ab} is positive.

3.1 The slab law

For a partition endpoint U, let $D(U)$ denote the union of slab states whose clock intervals lie in (U, ∞). Define the limiting late-clock mass

$$\overline{\nu}(U) = \int_{(0,1]} e^{-rU} \nu(\, dr). \tag{19}$$

By dominated convergence, $\overline{\nu}(U) \downarrow 0$.

Lemma 3.1 (Slab law and finite-state spectral gap)**.** *For every fixed partition,*

$$p_{n,a} \xrightarrow{\mathbb{P}} p_a, \qquad \frac{N_{n,ab}}{n} \xrightarrow{\mathbb{P}} q_{ab}$$

simultaneously in a, b. There are deterministic $\varepsilon_n \downarrow 0$ and events $\mathcal{G}_n$ with $\mathbb{P}(\mathcal{G}_n) \to 1$ such that, on $\mathcal{G}_n$:

(i) all these convergences have error at most ε_n, and every $m_{n,a}$ and $N_{n,ab}$ is bounded below by a positive constant times n;

(ii) for every fixed $q_0 \in (q(\nu), 1)$ and all sufficiently large n,

$$\|P_n\|_{L^2_0(p_n) \to L^2_0(p_n)} \le q_0; \tag{20}$$

(iii) if S is a partition endpoint and $J_b \subset [0, A(S)]$, then

$$P_{n,ab} \le 2 M_S p_{n,b} \quad \text{for every } a, \qquad M_S = \frac{1}{B(S)}; \tag{21}$$

(iv) for every partition endpoint U,

$$p_n(D(U)) \le \overline{\nu}(U) + \varepsilon_n. \tag{22}$$

Proof. For a fixed time cell $I_b = (u, v]$, the indicators $\mathbf{1}_{\{T_{n,i} \in I_b\}}$ are independent and have means $e^{-r_{n,i}u} - e^{-r_{n,i}v}$. Kolmogorov's maximal inequality gives

$$\max_{k \le n} \left| \sum_{i \le k} \left(\mathbf{1}_{\{T_{n,i} \in I_b\}} - \mathbb{E}\mathbf{1}_{\{T_{n,i} \in I_b\}} \right) \right| = o_{\mathbb{P}}(n).$$

Weak convergence of the marked empirical measures gives, uniformly in $x \in [0, 1]$,

$$\frac{1}{n} \sum_{i \le nx} \mathbb{E}\mathbf{1}_{\{T_{n,i} \in I_b\}} \longrightarrow \int_0^x \int (e^{-ru} - e^{-rv}) \eta_z(\, dr)\, dz.$$

This proves the slab-size limit and convergence of the endpoints of $Q_{n,a}/n$ to those of J_a. Evaluating the same uniform partial-sum limit at those random endpoints yields

$$\frac{N_{n,ab}}{n} \longrightarrow \int_{J_a} \int (e^{-rt_{b-1}} - e^{-rt_b}) \eta_x(\, dr)\, dx = q_{ab}.$$

The last equality follows by integrating (14) over J_b and using $y = A(t)$.

The matrix P is the compression of K_η to functions constant on the partition cells J_a. Conditional expectation is a contraction, so Lemma 2.2 gives $\|P\|_{L_0^2(p) \to L_0^2(p)} \le q(\nu)$. Entrywise convergence of the conjugated finite matrices $D_{p_n}^{1/2} P_n D_{p_n}^{-1/2}$ proves (20).

If $y \le A(S)$, then $\tau(y) \le S$ and

$$\rho(x, y) = \frac{g_x(\tau(y))}{B(\tau(y))} \le \frac{1}{B(S)} = M_S,$$

because $0 < r \le 1$ and B is decreasing. Hence $P_{ab} \le M_S p_b$, and entrywise convergence gives (21). Finally,

$$p_n(D(U)) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{T_{n,i} > U\}} \xrightarrow{\mathbb{P}} \int e^{-rU} \nu(\, dr) = \overline{\nu}(U).$$

Since the partition is finite, all statements can be placed on one sequence of events $\mathcal{G}_n$ with deterministic errors $\varepsilon_n \downarrow 0$. □

3.2 Conditional permanental matchings

Condition on the complete slab data: the sets $H_{n,a}$, the rank intervals $Q_{n,a}$, and the ordered but unlabeled clock values

$$s_{a,1} < \cdots < s_{a,m_{n,a}}$$

in every slab, including the tail slab. The tail row will never be assigned a flatness estimate; conditioning on its ordered values is used only for exact residual factorization. Write $Q_{n,a} = \{q_{a,1} < \cdots < q_{a,m_{n,a}}\}$ and define $\phi_a : H_{n,a} \to [m_{n,a}]$ by $\sigma_n(i) = q_{a,\phi_a(i)}$. For a bijection ϕ, independence of the exponential densities gives

$$\mathbb{P}(\phi_a = \phi \mid \text{slab data}) = \frac{\prod_{i \in H_{n,a}} e^{-r_{n,i} s_{a,\phi(i)}}}{\operatorname{per}(e^{-r_{n,i} s_{a,j}})_{i,j}}. \tag{23}$$

The row matchings are conditionally independent.

Lemma 3.2 (Residual permanental factorization)**.** *After conditioning on any compatible partial collection of exposed matching edges, delete the exposed source and target vertices in each row. The remaining row matchings are conditionally independent, and each has the permanental law* (23) *on its residual submatrix.*

Proof. The conditional density is a product over rows and, within a row, a product over matching edges. Conditioning on fixed edges removes their constant factors and restricts the normalization to bijections of the remaining rows and columns, whose sum is the corresponding permanent. □

Lemma 3.3 (Hereditary flatness)**.** *Let $A = (a_{ij})$ be a positive $m \times m$ matrix satisfying*

$$e^{-\epsilon} \le a_{ij}/a_{ik} \le e^{\epsilon}$$

for every row i and columns j, k. Under the permanental matching law proportional to $\prod_i a_{i,\phi(i)}$, after conditioning on an arbitrary partial matching, every remaining edge probability lies in

$$[e^{-2\epsilon}/s, e^{2\epsilon}/s],$$

where s rows and columns remain. For a clock slab of width at most h this holds with

$$\beta = 2h. \tag{24}$$

Proof. Work in the current residual submatrix. For two remaining columns j, k,

$$\frac{\mathbb{P}(\phi(i) = j)}{\mathbb{P}(\phi(i) = k)} = \frac{a_{ij} \operatorname{per} A_{-i,-j}}{a_{ik} \operatorname{per} A_{-i,-k}}.$$

Map each term of the first minor to a term of the second by locating the row sent to k and sending that row to j. Corresponding products differ by a factor in $[e^{-\epsilon}, e^{\epsilon}]$; the displayed prefactor contributes the same range. Thus pairwise edge-probability ratios lie in $[e^{-2\epsilon}, e^{2\epsilon}]$, and the individual bounds follow because the s probabilities sum to one. In a slab, $e^{-h} \le e^{-r_{n,i}(s_j - s_k)} \le e^{h}$. □

Lemma 3.4 (Predictable exposure)**.** *Suppose that before each new edge is exposed, its source is measurable with respect to the slab data and all previously exposed edges. Conditional on the current filtration, its target has the one-edge marginal of the residual permanental matching. If s targets remain in a finite slab, each target has conditional probability in*

$$[e^{-\beta}/s, e^{\beta}/s].$$

If one distinguished target is forbidden, each of the other $s - 1$ targets has conditional probability in $[e^{-\beta}/(s-1), e^{\beta}/(s-1)]$.

Proof. Apply Lemma 3.2 before the new edge is revealed and then Lemma 3.3. Conditioning away from a distinguished target renormalizes the other probabilities without changing their pairwise ratios. □

3.3 A universal contraction for late rows

The following elementary inequality replaces every global Hilbert–Schmidt estimate.

Lemma 3.5 (Markov contraction of a row block)**.** *Let P be a finite stochastic matrix with positive stationary probability vector p. For any set D of row states, put*

$$s_b = \sum_{a \in D} p_a P_{ab}.$$

Then

$$\sum_b \frac{s_b^2}{p_b} \le p(D). \tag{25}$$

Proof. Stationarity gives $0 \le s_b \le p_b$, and stochasticity gives $\sum_b s_b = p(D)$. Therefore

$$\sum_b \frac{s_b^2}{p_b} \le \sum_b s_b = p(D).$$

□

Remark 3.6. The estimate is exact at finite n, requires no reversibility, and remains valid when the limiting kernel is not Hilbert–Schmidt. For the ordered Sukhatme coupling $r(x) = x$, with $\nu(\,dr) = \,dr$ on $(0,1)$, the squared Hilbert–Schmidt integral truncated at clock time T satisfies

$$\begin{aligned}\int_0^1 \int_0^{A(T)} \rho(x,y)^2 \,dy\,dx &= \int_0^T \frac{\int_0^1 r^2 e^{-2rt}\,dr}{B_\nu(t)}\,dt \\ &= \frac{1}{4}\log T + O(1), \qquad T \to \infty.\end{aligned}$$

Indeed, $B_\nu(t) = (1-(1+t)e^{-t})/t^2$, and the integrand in the time integral is $1/(4t) + O(e^{-t})$ as $t \to \infty$. The decreasing ordered coupling gives the same integral by reflection. Thus the limiting kernel is not Hilbert–Schmidt, while Lemma 3.5 still controls late rows by the square root of their total mass.

3.4 A relative-prefix estimate

We use the following adaptive sampling lemma. It is included to make the regeneration argument self-contained.

Lemma 3.7 (Relative prefix law)**.** *Fix $d \ge 1$ and $c_* > 0$. There are absolute constants $\beta_0 > 0$, $C < \infty$, and deterministic sequences $\chi_m \downarrow 0$, $\pi_m \downarrow 0$, depending only on d, c_*, with the following property. Partition m targets into d categories of sizes $m_1, \dots, m_d$, where $m_b \ge c_* m$. Select targets sequentially without replacement in an arbitrary predictable adaptive source order. Suppose that, whenever s targets remain, every individual remaining target has conditional selection probability in*

$$[e^{-\beta}/s, e^{\beta}/s], \qquad 0 \le \beta \le \beta_0.$$

If $X_b(r)$ is the number of category-b targets among the first r selections, then

$$\mathbb{P}\left(\max_{b \le d} \max_{0 \le r \le m} \frac{|X_b(r) - r m_b/m|}{m_b} > C\beta + \chi_m\right) \le \pi_m. \tag{26}$$

The conclusion is unchanged, after harmless modification of the constants, if a fixed number of distinguished targets is deleted or forbidden.

Proof. Fix one category and write

$$N_r = m_b - X_b(r), \qquad s_r = m - r.$$

If Y_{r+1} indicates that the next target lies in this category, then

$$\mathbb{E}(Y_{r+1} \mid \mathscr{F}_r) \in \left[e^{-\beta}\frac{N_r}{s_r}, e^{\beta}\frac{N_r}{s_r}\right]. \tag{27}$$

Set $a_m = m_b^{2/3}$ and stop when $N_r < a_m$ or $s_r < a_m$. Before that time put $Z_r = \log(N_r/s_r)$. Taylor expansion gives

$$|\mathbb{E}(Z_{r+1} - Z_r \mid \mathscr{F}_r)| \le C\frac{\beta}{s_r} + C\left(\frac{1}{s_r^2} + \frac{1}{s_r N_r}\right),$$

and the conditional variance of the centered increment is at most $C/(s_r N_r)$. Since $m_b \ge c_* m$,

$$\sum_{r<\tau} \frac{1}{s_r N_r} \le C_{c_*}\frac{1+\log m}{m^{2/3}} =: v_m.$$

Doob's L^2 maximal inequality shows that, outside an event of probability $O(v_m^{1/2})$, the stopped martingale part is uniformly $O(v_m^{1/4})$. Summing the drift and writing $u = s_r/m$ yields

$$e^{-\eta_m} u^{1+C\beta} \le \frac{N_r}{m_b} \le e^{\eta_m} u^{1-C\beta}, \qquad \eta_m = O(v_m^{1/4} + v_m).$$

Choose β_0 so that $C\beta_0 \le 1/2$. The elementary bound

$$\sup_{0\le u\le 1} |u^{1+a} - u| \le C|a|, \qquad |a| \le 1/2,$$

then gives

$$\left|\frac{N_r}{m_b} - \frac{s_r}{m}\right| \le C\beta + o(1)$$

uniformly before the stopping time.

If $s_\tau < a_m$, the upper power bound forces $N_\tau/m_b = o(1)$. If instead $N_\tau < a_m$, the lower power bound forces $s_\tau/m = o(1)$. Both normalized remaining counts are nonincreasing after τ, so the same estimate holds for all later prefixes. A union bound over the fixed number of categories proves (26). Deleting or forbidding finitely many targets changes all relevant counts by $O(1)$ only. □

In all later applications the fixed mesh is chosen so that $\beta = 2h \le \beta_0$.

4 Scale-sensitive orbit balance

A residual history $\mathcal{H}$ consists of the slab data, all roots selected so far, and the matching edges in a collection of fully exposed and deleted cycles; it contains no additional information about unexposed row matchings. Let $R \subset [n]$ be the remaining vertex set, $N = |R|$, and $r = N/n$. Put

$$m_a^R = |H_{n,a} \cap R|, \qquad N_{ab}^R = |Q_{n,a} \cap H_{n,b} \cap R|.$$

For every deleted cycle C and every slab a,

$$|C \cap H_{n,a}| = |C \cap Q_{n,a}|, \tag{28}$$

because σ_n maps the first set bijectively onto the second. Hence $m_a^R = |Q_{n,a} \cap R|$.

Fix a partition endpoint U. A history is (U, δ, ξ)-regular if $N \geq \xi n$, $0 \leq \delta < 1/4$, and, for every row contained in $[0, U]$ and every target state b,

$$\left| \frac{m_a^R}{r m_{n,a}} - 1 \right| \leq \delta, \qquad \left| \frac{N_{ab}^R}{r N_{n,ab}} - 1 \right| \leq \delta. \tag{29}$$

The initial history is $(U, 0, 1)$-regular.

An admissible exploration reveals one outgoing edge at a time, with the next source measurable before the edge is exposed. We allow a fixed number of distinguished targets to be forbidden. Lemma 3.4 applies to every actual cycle exploration and every root-avoiding exploration below.

Consider an injective path segment $v_0, \ldots, v_\ell$, allowing $v_\ell = v_0$ only at a terminal closing step. Put

$$R_a(\ell) = \#\{0 \leq j < \ell : v_j \in H_{n,a}\},$$

and

$$R_{ab}(\ell) = \#\{0 \leq j < \ell : v_j \in H_{n,a},\ v_{j+1} \in H_{n,b}\}.$$

The deterministic flow identity is

$$R_b(\ell) - \sum_a R_{ab}(\ell) = \mathbf{1}_{\{v_0 \in H_{n,b}\}} - \mathbf{1}_{\{v_\ell \in H_{n,b}\}}. \tag{30}$$

Lemma 4.1 (Scale-sensitive orbit balance)**.** *Fix partition endpoints $0 < S < U \leq T$, $\xi > 0$, and $0 \leq \delta < 1/4$. There are deterministic $\chi_n, \pi_n \downarrow 0$ such that, on $\mathcal{G}_n$, uniformly over every (U, δ, ξ)-regular history and every admissible exploration, with conditional probability at least $1 - \pi_n$ the following statements hold simultaneously at every defined prefix length $\ell \geq 1$.*

For every row a contained in $[0, U]$ and every target state b,

$$\left| R_{ab}(\ell) - \frac{R_a(\ell)}{m_a^R} N_{ab}^R \right| \leq (C\beta + \chi_n) N_{ab}^R. \tag{31}$$

Define

$$\Delta_n(U, \delta, h) = \delta + \beta + \chi_n + \sqrt{p_n(D(U))}. \tag{32}$$

Then, for every inner state b with $J_b \subset [0, A(S)]$,

$$\left| \frac{R_b(\ell)}{\ell p_{n,b}} - 1 \right| \leq C_{S,\xi} \frac{n}{\ell} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J}}}{\ell}. \tag{33}$$

In particular, for every fixed $\lambda > 0$, the right-hand side is $C_{S,\xi,\lambda} \Delta_n + o(1)$ uniformly for $\ell \geq \lambda n$.

Proof. On $\mathcal{G}_n$, every cell in the fixed finite partition has size bounded below by a positive constant times n. Regularity gives the same bound for the residual cells. Apply Lemmas 3.4 and 3.7 row by row, extending the queried source sequence predictably after the orbit ends if necessary. A finite union bound gives (31) with sequences χ_n, π_n uniform over all stated histories and explorations.

Suppress ℓ and put

$$P = P_n, \qquad q_a = R_a/\ell, \qquad E_{ab} = R_{ab} - R_a P_{ab}.$$

For $a \notin D(U)$, regularity and (31) give

$$|E_{ab}| \le C(\delta + \beta + \chi_n) N_{n,ab}. \tag{34}$$

For $a \in D(U)$, injectivity gives $0 \le R_{ab} \le N_{n,ab}$ and $0 \le R_a P_{ab} \le N_{n,ab}$, hence

$$|E_{ab}| \le N_{n,ab}. \tag{35}$$

Let

$$G_b = \sum_{a \notin D(U)} E_{ab}, \qquad H_b = \sum_{a \in D(U)} E_{ab}.$$

Since $\sum_a N_{n,ab} = n p_{n,b}$,

$$\|G/(\ell p_n)\|_{L^2(p_n)} \le C \frac{n}{\ell} (\delta + \beta + \chi_n). \tag{36}$$

For the bad rows, (35) and Lemma 3.5 give

$$\sum_b \frac{H_b^2}{m_{n,b}} \le n \sum_b \frac{\left(\sum_{a \in D(U)} p_{n,a} P_{n,ab}\right)^2}{p_{n,b}} \le n p_n(D(U)). \tag{37}$$

Consequently

$$\|H/(\ell p_n)\|_{L^2(p_n)} \le \frac{n}{\ell} \sqrt{p_n(D(U))}. \tag{38}$$

Let d_b be the right-hand side of (30); it has at most two nonzero coordinates of absolute value at most one. Dividing the flow identity by ℓ gives

$$q - qP = e, \qquad e_b = (d_b + G_b + H_b)/\ell.$$

Put $u_a = q_a / p_{n,a} - 1$. Since $p_n P = p_n$ and $\sum_a p_{n,a} u_a = 0$,

$$u - P^* u = e / p_n. \tag{39}$$

By (20),

$$\|u\|_{L^2(p_n)} \le (1 - q_0)^{-1} \, \|e/p_n\|_{L^2(p_n)} \le C \frac{n}{\ell} \Delta_n + \frac{C_{\mathcal{J}}}{\ell}. \tag{40}$$

For an inner state b, (21) and stationarity imply

$$\sum_a p_{n,a} \left(\frac{P_{n,ab}}{p_{n,b}} \right)^2 \le 2 M_S.$$

Thus $|(P^* u)_b| \le \sqrt{2 M_S} \, \|u\|_{L^2(p_n)}$. Directly from (34), (35), and (21),

$$\left| \frac{e_b}{p_{n,b}} \right| \le C_S \frac{n}{\ell} (\delta + \beta + \chi_n + p_n(D(U))) + \frac{C_{\mathcal{J}}}{\ell}.$$

Now use $u_b = (e/p_n)_b + (P^* u)_b$ and $p_n(D(U)) \le \sqrt{p_n(D(U))}$ to obtain (33). □

5 Rooted cycles and regeneration

For a vertex v, let $c(v)$ be the row index for which $v \in Q_{n,c(v)}$. An orbit can hit v only when its current source lies in $H_{n,c(v)}$.

5.1 Root avoidance

Lemma 5.1 (Root-avoiding change of measure)**.** *Fix a residual history $\mathcal{H}$ and a distinguished remaining target v. Explore the path from v. At each step, use the original residual one-edge kernel conditional on the complete exposed history. At its jth opportunity to hit v, indexed from zero, let λ_j be that kernel's probability of selecting v. Define the auxiliary law $\mathbb{P}^\circ$ by conditioning each such transition away from v, and stopping if no allowed target remains.*

Let τ° be the number of nonclosing edges completed before this stopping time. On $\{\tau^\circ \geq k\}$, let $S(k)$ be the number of closure opportunities among the first k completed edges, and define

$$W_k = \begin{cases} \displaystyle\prod_{j=0}^{S(k)-1} (1-\lambda_j), & \tau^\circ \geq k, \\ 0, & \tau^\circ < k. \end{cases} \tag{41}$$

Then

$$\mathbb{P}(|\mathcal{C}(v)| > k \mid \mathcal{H}, v) = \mathbb{E}^\circ W_k. \tag{42}$$

If the closure row $c(v)$ is a finite clock slab of width at most h and initially has m available targets, including v, then, while defined,

$$\frac{e^{-\beta}}{m-j} \leq \lambda_j \leq \frac{e^{\beta}}{m-j}. \tag{43}$$

Proof. At a closure opportunity with $\lambda_j < 1$, the original transition law is a mixture of an atom of mass λ_j at v and the avoid-v law of mass $1-\lambda_j$. Multiplying these sequential Radon–Nikodym factors along each auxiliary prefix of length k and summing over prefixes proves (42). Histories stopped before completing k edges contribute zero. A previously visited non-root vertex cannot be the first repeat because its target column was consumed when the path first entered it. Before the jth opportunity, exactly j non-root targets have been removed from the closure row. When that row has width at most h, Lemma 3.4 gives (43). □

Lemma 5.2 (Harmonic product)**.** *Fix $\zeta \in (0,1)$. Uniformly for $c \in [1/2, 2]$, integers $m \geq 1$, and $0 \leq s \leq (1-\zeta)m$,*

$$\prod_{j=0}^{s-1} \left(1 - \frac{c}{m-j}\right) = (1-s/m)^c\{1 + O_\zeta(m^{-1})\}, \tag{44}$$

whenever all factors are nonnegative. Also, uniformly for $x \in [\zeta, 1-\zeta]$ and $|a| \leq 1/2$,

$$|(1-x)^{1+a} - (1-x)| \leq C_\zeta |a|.$$

Proof. Expand the logarithm. Since $m - j \geq \zeta m$,

$$\sum_{j<s} \log\left(1 - \frac{c}{m-j}\right) = -c \sum_{j<s} \frac{1}{m-j} + O_\zeta(m^{-1}),$$

and the harmonic sum equals $\log(m/(m-s)) + O_\zeta(m^{-1})$. The second assertion follows from the mean-value theorem in the exponent. □

5.2 A uniform residual fraction

Proposition 5.3 (Uniform conditional residual fraction)**.** *Fix partition endpoints $0 < S < U \le T$, $\xi > 0$, $\zeta \in (0, 1/2)$, and $0 \le \delta < 1/4$. On $\mathcal{G}_n$, let $\mathcal{H}$ be a (U, δ, ξ)-regular history. Choose a root uniformly from its N remaining vertices, independently of the unexposed matchings, and let L be the root-cycle length. Uniformly over all such histories,*

$$\sup_{x\in[\zeta,1-\zeta]} \left| \mathbb{P}\left(\frac{L}{N} \le x \,\middle|\, \mathcal{H} \right) - x \right| \le C_{S,\xi,\zeta}\Delta_n(U,\delta,h) + C_\xi p_n(D(S)) + \pi_n + \frac{C_{\mathcal{J},\xi,\zeta}}{n}. \tag{45}$$

Proof. Let v be the uniform root and $C = c(v)$. By (28),

$$\mathbb{P}(C = a \mid \mathcal{H}) = \frac{|Q_{n,a} \cap R|}{N} = \frac{m_a^R}{N}.$$

Therefore

$$\mathbb{P}(C \in D(S) \mid \mathcal{H}) \le \frac{np_n(D(S))}{N} \le \xi^{-1} p_n(D(S)). \tag{46}$$

Work henceforth on $C \notin D(S)$. Put $m = m_C^R$ and $p_C = p_{n,C}$. Regularity gives

$$\frac{m}{Np_C} = 1 + O(\delta). \tag{47}$$

For the fixed partition, p_C is bounded below over inner rows, so $m \ge c_{\mathcal{J},S}\xi n$.

First exclude premature termination of the avoid-v exploration. If it stops after r edges because only v remains in the closure row, then $R_C(r) = m - 1$. On the simultaneous event of Lemma 4.1,

$$\left| \frac{m-1}{rp_C} - 1 \right| \le C_S \frac{n}{r}\Delta_n + \frac{C_{\mathcal{J}}}{r}.$$

Multiplying by r/N and using (47) gives

$$|1 - r/N| \le C_{S,\xi}\Delta_n + O_{\mathcal{J},\xi}(n^{-1}). \tag{48}$$

Unless the desired bound is already trivial, this excludes termination before $(1-\zeta)N$ edges.

Fix $x \in [\zeta, 1-\zeta]$ and $k = \lfloor xN \rfloor$. On the simultaneous event above, $S(k) = R_C(k)$. Lemma 4.1 and (47) give

$$\sup_{x\in[\zeta,1-\zeta]} \left| \frac{S(\lfloor xN \rfloor)}{m} - x \right| \le C_{S,\xi,\zeta}\Delta_n + O_{\mathcal{J},\xi,\zeta}(n^{-1}) \tag{49}$$

except on conditional probability at most π_n.

On this event, the weight W_k in Lemma 5.1 is squeezed between the two harmonic products with $c = e^\beta$ and $c = e^{-\beta}$. Since the mesh is chosen with β small, Lemma 5.2 and (49) show that both differ from $1 - x$ by the principal error in (45). Average under $\mathbb{P}^\circ$, use the trivial bound on the exceptional event, and add (46). □

5.3 Deletion stability

Lemma 5.4 (Deletion of an interior macroscopic cycle)**.** *Fix partition endpoints $0 < V < U \le T$, $\xi > 0$, $\zeta \in (0, 1/2)$, and $0 \le \delta < 1/4$. On $\mathcal{G}_n$, start from a (U, δ, ξ)-regular history and expose the cycle of a uniform residual root. Let N and L be the residual size and cycle length. There is a constant $C = C(V, \xi, \zeta)$ such that*

$$\mathbb{P}\Big(\zeta N \le L \le (1-\zeta)N,\ \textit{the post-deletion history is not } (V, \delta', \zeta\xi)\textit{-regular} \mid \mathcal{H}\Big) \le \pi_n, \tag{50}$$

where

$$\delta' = C\Delta_n(U, \delta, h) + \frac{C_{\mathcal{J},\xi,\zeta}}{n}. \tag{51}$$

The assertion is used only when $\delta' < 1/4$.

Proof. Let d_a be the number of vertices of the exposed cycle in $H_{n,a}$ and d_{ab} the number of its edges from $H_{n,a}$ to $H_{n,b}$. The event in Lemma 4.1 is simultaneous over all prefix lengths and may therefore be evaluated at the random closing time without conditioning on L. On $\zeta N \le L \le (1-\zeta)N$, apply (33) with inner cutoff V and $\ell = L$ to get

$$\frac{d_a}{L p_{n,a}} = 1 + O_{V,\xi,\zeta}(\Delta_n) + O_{\mathcal{J},\xi,\zeta}(n^{-1})$$

for every row below V. Since $m_a^R = N p_{n,a}(1 + O(\delta))$,

$$\frac{d_a}{m_a^R} = \frac{L}{N} + O_{V,\xi,\zeta}(\Delta_n) + O_{\mathcal{J},\xi,\zeta}(n^{-1}). \tag{52}$$

The row-prefix estimate at the same closing time gives

$$d_{ab} = \frac{d_a}{m_a^R} N_{ab}^R + O((\beta + \chi_n) N_{ab}^R),$$

so

$$d_{ab} = \frac{L}{N} N_{ab}^R + \{O_{V,\xi,\zeta}(\Delta_n) + O_{\mathcal{J},\xi,\zeta}(n^{-1})\} N_{ab}^R. \tag{53}$$

The quantity d_{ab} is exactly the number of targets removed from $Q_{n,a} \cap H_{n,b}$. Equations (52)–(53), together with $1 - L/N \ge \zeta$, prove regularity of the residual history with error (51). Residual factorization remains valid by Lemma 3.2. □

6 The regeneration induction

Define residual fractions

$$V_{n,j} = \frac{\widetilde{L}_{n,j}}{n - \widetilde{L}_{n,1} - \cdots - \widetilde{L}_{n,j-1}}, \tag{54}$$

with $V_{n,j} = 0$ after exhaustion.

Lemma 6.1 (Conditional CDF test)**.** *Let $W \in [0, 1]$, let $\mathscr{F}$ be a sigma-field, and suppose*

$$\sup_{x \in [a,b]} |\mathbb{P}(W \le x \mid \mathscr{F}) - x| \le \eta \quad a.s., \qquad 0 < a < b < 1.$$

Then

$$\mathbb{P}(W \notin [a,b] \mid \mathscr{F}) \le a + (1-b) + 2\eta.$$

If f is absolutely continuous, supported in $[a,b]$, and vanishes at the endpoints, then

$$\left|\mathbb{E}(f(W) \mid \mathscr{F}) - \int_0^1 f(u)\,du\right| \le \eta \int_a^b |f'(u)|\,du.$$

Proof. The endpoint estimate follows by evaluating the CDF bound at a, b. The second claim is conditional Stieltjes integration by parts. □

Proposition 6.2 (Independent residual uniforms)**.** *For every fixed k,*

$$(V_{n,1}, \dots, V_{n,k}) \Rightarrow (U_1, \dots, U_k), \qquad U_j \overset{\text{iid}}{\sim} \text{Unif}(0,1). \tag{55}$$

Proof. Fix k, an interior parameter $\zeta \in (0, 1/4)$, and a target error $\eta \in (0, 1/16)$. Put $\xi_* = \zeta^{k-1}$. We describe the order of choices, which is essential.

Choose an inner cutoff T_0 so large that $\xi_*^{-1}\overline{\nu}(T_0) \le \eta/4$. Once T_j is fixed, choose constants $R_j, D_j \ge 1$ bounding the principal-error coefficients in, respectively, Proposition 5.3 and Lemma 5.4, with inner cutoff T_j, residual fraction at least ξ_*, and truncation parameter ζ. These constants are independent of the outer cutoff and of the subsequent partition. Set

$$A_j = \prod_{i=0}^{j} (1 + R_i + D_i),$$

and choose $T_{j+1} > T_j$ so large that

$$A_j \sqrt{\overline{\nu}(T_{j+1})} \le 2^{-j-3}\eta.$$

Continue through T_k. This is possible because $\overline{\nu}(U) \downarrow 0$.

Write $\epsilon_{j+1} = \sqrt{\overline{\nu}(T_{j+1})}$. An error introduced at level j contributes to a later regularity error at level $i \le j$ with coefficient $\prod_{\ell=i}^{j} D_\ell$, and to the root-law error at level i with coefficient $R_i \prod_{\ell=i+1}^{j} D_\ell$, with empty products equal to one. Both coefficients are at most A_j. Thus the total tail contribution to any regularity or root-law error is at most

$$\sum_{j=0}^{k-1} A_j \epsilon_{j+1} \le \eta/4.$$

After all cutoffs and constants are fixed, choose $h > 0$ such that

$$2h \le \beta_0, \qquad 2h \sum_{j=0}^{k-1} A_j \le \eta/4.$$

Choose one finite partition of $[0, T_k]$ containing $T_0, \dots, T_k$ and having mesh at most h, and append its tail cell. The same propagation calculation bounds every mesh contribution by $\eta/4$. Finally take n large enough that the fixed-partition errors, the empirical tail-mass errors, and the exceptional-event probabilities, after these finitely many amplifications, contribute at most another prescribed fraction of η. Every required regularity error is then below $1/4$, and every

stagewise conditional CDF error is $O_k(\eta)$ without a cutoff-dependent factor multiplying η. For reference, at deletion stage $1 \le r \le k$ we use

stage	current outer cutoff	root-law inner cutoff	post-deletion cutoff
r	T_{k-r+1}	T_{k-r}	T_{k-r}.

Thus every inner cutoff is fixed before its corresponding outer cutoff.

On the event

$$\zeta \le V_{n,j} \le 1-\zeta, \qquad 1 \le j < r, \tag{56}$$

the residual size before stage r is at least $\zeta^{r-1}n \ge \xi_* n$. Inductively, Lemma 5.4 keeps the current history regular at the required outer cutoff, apart from a conditional probability $O_k(\eta)$. Proposition 5.3 then gives

$$\sup_{x\in[\zeta,1-\zeta]} |\mathbb{P}(V_{n,r} \le x \mid \mathscr{F}_{r-1}) - x| \le C_k\eta \tag{57}$$

uniformly over every surviving regular history.

Let $f_1,\dots,f_k$ be bounded Lipschitz functions on $[0,1]$. Choose a piecewise linear cutoff φ_ζ supported in $[\zeta,1-\zeta]$ and equal to one on $[2\zeta,1-2\zeta]$, and put $g_j = f_j\varphi_\zeta$. Apply Lemma 6.1 to (57), multiply successively by the preceding factors, and use the joint-event form of Lemma 5.4. Induction gives

$$\mathbb{E}\prod_{j=1}^{k} g_j(V_{n,j}) = \prod_{j=1}^{k}\int_0^1 g_j(u)\,du + O_{f,k,\zeta}(\eta) + o_n(1).$$

The endpoint part of Lemma 6.1 gives

$$\limsup_{n\to\infty} \mathbb{P}\left(V_{n,j} \notin [2\zeta,1-2\zeta] \text{ for some } j \le k\right) \le 4k\zeta + C_k\eta.$$

Replace g_j by f_j, then let $n\to\infty$, $\eta\downarrow 0$, and $\zeta\downarrow 0$. Products of bounded Lipschitz functions determine weak convergence on $[0,1]^k$, proving (55). □

Proof of Theorem 1.1. Put

$$X_{n,j} = \frac{\widetilde{L}_{n,j}}{n} = V_{n,j}\prod_{i<j}(1-V_{n,i}), \qquad R_{n,k} = 1 - \sum_{j=1}^{k} X_{n,j} = \prod_{i=1}^{k}(1-V_{n,i}). \tag{58}$$

Proposition 6.2 and the continuous mapping theorem give for every fixed k the corresponding finite-dimensional GEM(1) limit, including

$$R_{n,k} \Rightarrow \prod_{i=1}^{k}(1-U_i).$$

Since $0 \le R_{n,k} \le 1$,

$$\mathbb{E}R_{n,k} \longrightarrow 2^{-k}. \tag{59}$$

Let $X_n = (X_{n,1}, X_{n,2}, \dots)$ and let X have the GEM(1) law. If $F : \ell^1 \to \mathbb{R}$ is bounded and Lipschitz, truncation after k coordinates gives

$$|\mathbb{E}F(X_n) - \mathbb{E}F(X)| \le \mathrm{Lip}(F)(\mathbb{E}R_{n,k} + 2^{-k}) + |\mathbb{E}F(X_n^{(k)}) - \mathbb{E}F(X^{(k)})|.$$

First let $n \to \infty$ and then $k \to \infty$. The bounded-Lipschitz criterion on the separable space ℓ^1 proves (6).

For nonnegative ℓ^1 sequences, decreasing rearrangement is a contraction:

$$\left\| x^{\downarrow} - y^{\downarrow} \right\|_1 \le \|x - y\|_1 .$$

This follows first for finite sequences by the two-point uncrossing inequality and then by truncation. The continuous mapping theorem proves (7). The first residual fraction is the cycle length of a uniform root divided by n, proving (8).

The argument was carried out along an arbitrary subsequence after extracting a marked limit. Lemma 2.2 gives the same strict bound $q(\nu)$ for every such marked limit, and the resulting law is always universal. The subsequence principle therefore yields convergence along the full sequence. □

Corollary 6.3 (Power sums)**.** *Under the hypotheses of Theorem 1.1, for every $s > 1$,*

$$\sum_{j \ge 1} \left(\frac{L_{n,j}}{n} \right)^s \Rightarrow \sum_{j \ge 1} P_j^s, \qquad (P_j) \sim \mathrm{PD}(1),$$

and

$$\mathbb{E} \sum_{j \ge 1} \left(\frac{L_{n,j}}{n} \right)^s \longrightarrow \frac{1}{s}.$$

Proof. The power-sum functional is bounded and continuous on $\mathcal{S}^{\downarrow}$. A size-biased pick $\widetilde{P}$ from a PD(1) partition is uniform on $(0,1)$, and

$$\mathbb{E}(\widetilde{P}^{s-1} \mid (P_j)) = \sum_j P_j^s.$$

Hence the limiting expectation is $\int_0^1 u^{s-1}\, du = 1/s$. □

7 Regularly varying spectral gaps

We prove Theorem 1.5. Let

$$F(x) = \nu((0,x])$$

be regularly varying at zero with index $\alpha > 0$. Define the truncated first moment

$$H(x) = \int_{(0,x]} r\nu(\, dr). \tag{60}$$

Integration by parts and Karamata's theorem give

$$H(x) = xF(x) - \int_0^x F(u)\, du \sim \frac{\alpha}{\alpha+1} xF(x) \qquad (x \downarrow 0). \tag{61}$$

Thus H is regularly varying at zero with index $\alpha + 1$. Since $B_\nu(t) = \int e^{-tr}\, dH(r)$, the Abelian theorem for Laplace–Stieltjes transforms yields

$$B_\nu(t) \sim \Gamma(\alpha+2) H(1/t) \sim \alpha\Gamma(\alpha+1) t^{-1} F(1/t) \qquad (t \to \infty). \tag{62}$$

These standard forms of Karamata's theorem are recorded in [2, Chapters 1 and 4].

Let $C = C_\nu$ and $Q = CC^*$ on $L^2(\nu)$. A direct calculation gives the symmetric Markov kernel

$$R_\nu(r,s) = rs \int_0^\infty \frac{e^{-(r+s)t}}{B_\nu(t)}\, dt \tag{63}$$

relative to $\nu(\,ds)$.

Lemma 7.1 (Dyadic kernel lower bound)**.** *Let*

$$I_k = (2^{-k-1}, 2^{-k}], \qquad w_k = \nu(I_k).$$

There are $K < \infty$ and $c > 0$ such that, for every $k \geq K$, if r, s lie in the same shell I_k or in the adjacent shells I_k, I_{k-1}, then

$$R_\nu(r,s) \geq \frac{c}{w_k}. \tag{64}$$

Moreover,

$$\frac{w_{k+1}}{w_k} \longrightarrow 2^{-\alpha}. \tag{65}$$

Proof. Put $x = 2^{-k}$. Regular variation gives

$$w_k = F(x) - F(x/2) \sim (1 - 2^{-\alpha})F(x),$$

which proves (65). By (62) and the uniform convergence theorem, uniformly for $t \in [x^{-1}, 2x^{-1}]$ and all sufficiently small x,

$$B_\nu(t) \leq CxF(x). \tag{66}$$

If r, s are in the stated shells, then $rs \geq c_1 x^2$ and $r + s \leq c_2 x$. Restricting (63) to $t \in [x^{-1}, 2x^{-1}]$ therefore gives

$$R_\nu(r,s) \geq cx^2 \int_{x^{-1}}^{2x^{-1}} \frac{dt}{xF(x)} \geq \frac{c'}{F(x)} \geq \frac{c''}{w_k}.$$

□

We also need a weighted Poincaré inequality for an eventually geometric sequence of shell masses.

Lemma 7.2 (Geometric weighted Poincaré inequality)**.** *Let $(w_k)_{k \geq K}$ be positive and suppose that $w_{k+1}/w_k \leq \rho < 1$ for all $k \geq K$. For a sequence (m_k),*

$$\sum_{k \geq K} w_k (m_k - m_K)^2 \leq C_\rho \sum_{k > K} w_k (m_k - m_{k-1})^2. \tag{67}$$

Consequently the same right-hand side controls the weighted variance of $(m_k)_{k \geq K}$.

Proof. Put $d_j = m_j - m_{j-1}$ and $a_j = \sqrt{w_j} d_j$. For $j \leq k$, $w_k / w_j \leq \rho^{k-j}$, and hence

$$\sqrt{w_k}(m_k - m_K) = \sum_{j=K+1}^{k} \sqrt{\frac{w_k}{w_j}}\, a_j.$$

Young's convolution inequality on ℓ^2, with the summable kernel $(\rho^{r/2})_{r \geq 0}$, proves (67). Weighted variance is no larger than squared distance from the fixed value m_K. □

Proof of Theorem 1.5. Choose K so that Lemma 7.1 holds and $w_{k+1}/w_k \le \rho < 1$ for $k \ge K$. Put $I_\circ = (2^{-K}, 1]$ and $w_\circ = \nu(I_\circ)$. For K large, $w_\circ > 0$ and $w_K > 0$. Restricting (63) to $t \in [0,1]$, and using $B_\nu(t) \le 1$, gives a positive lower bound for R_ν on $[2^{-K-1}, 1]^2$. In particular, the Dirichlet form controls both the variance inside the core and the squared difference between the core mean and the mean on I_K.

Because Q is a symmetric Markov operator with kernel R_ν relative to ν, its Dirichlet form is

$$\langle f, (I-Q)f\rangle = \frac{1}{2}\iint (f(r)-f(s))^2 R_\nu(r,s)\,\nu(\,dr)\nu(\,ds). \tag{68}$$

Let $m_\circ$ be the ν-average of f on $I_\circ$ and let m_k be its average on I_k. The same-shell part of (64) yields

$$\langle f, (I-Q)f\rangle \ge c\left\{\int_{I_\circ} |f-m_\circ|^2\,d\nu + \sum_{k\ge K}\int_{I_k} |f-m_k|^2\,d\nu\right\}, \tag{69}$$

while the core–first-shell and adjacent-shell parts, followed by Jensen's inequality, give

$$\langle f, (I-Q)f\rangle \ge c\left\{w_K(m_K-m_\circ)^2 + \sum_{k>K} w_k(m_k-m_{k-1})^2\right\}. \tag{70}$$

Since $\sum_{k\ge K} w_k \le w_K/(1-\rho)$, Lemma 7.2 gives

$$\sum_{k\ge K} w_k(m_k-m_\circ)^2 \le 2C_\rho \sum_{k>K} w_k(m_k-m_{k-1})^2 + \frac{2w_K}{1-\rho}(m_K-m_\circ)^2. \tag{71}$$

Combining this with (69)–(70) and $\mathrm{Var}_\nu(f) \le \int |f-m_\circ|^2\,d\nu$ yields

$$\mathrm{Var}_\nu(f) \le C_\nu\langle f, (I-Q)f\rangle. \tag{72}$$

Since $Q = CC^*$ is positive semidefinite, (72) implies $\|Q\|_{L_0^2(\nu)} \le 1-c_\nu$ for some $c_\nu > 0$. Therefore

$$q(\nu) = \|C\|_{L_0^2(\mu_\nu)\to L_0^2(\nu)} \le \sqrt{1-c_\nu} < 1.$$

The power-law law (10) has $F(x) = x^\alpha$, so it is a special case. □

Remark 7.3. The proof gives no sharp value of $q(\nu)$. Even for the exact power-law law, determining the optimal gap appears to be a nontrivial Mellin-analytic problem.

8 Applications

8.1 Comparable arrays

Corollary 8.1 (Comparable rates)**.** *Suppose that after common rescaling all rates lie in* $[c,1]$ *for some fixed* $c > 0$*. Then the conclusions of Theorem 1.1 hold.*

Proof. For every $r \in [c,1]$,

$$re^{-rt}\,dt \ge ce^{-t}\,dt.$$

Thus all conditional time laws contain the common probability component $e^{-t}\,dt$ with mass c. The common-component decomposition and Jensen's inequality give $q(\nu) \le \sqrt{1-c} < 1$ for every subsequential empirical rate law ν. Although the full sequence of empirical laws need not converge, every subsequence has a further convergent subsequence, and Theorem 1.1 gives the same universal limit on each such refinement. The subsequence principle proves the full conclusion. □

8.2 Power-law and Sukhatme spectra

Proof of Corollary 1.6. Normalize by n^γ. The empirical measure of $(j/n)^\gamma$, $1 \le j \le n$, converges to the pushforward of Lebesgue measure under $x \mapsto x^\gamma$. Its density is

$$\frac{1}{\gamma} r^{1/\gamma - 1}\, dr = \nu_{1/\gamma}(\, dr).$$

The assignment of this multiset to labels does not change the empirical rate law. Apply Theorems 1.1 and 1.5. The Sukhatme case is $\gamma = 1$. □

The ordered Sukhatme profile $n, n-1, \dots, 1$ also has a distinct microscopic terminal theory: its fixed points have logarithmic intensity and a terminal Poisson process; see [12] and the historical exponential spacing representations of Sukhatme [18] and Rényi [16]. The present macroscopic result is insensitive to ordering and therefore belongs to a different universality mechanism.

8.3 Regularly varying spectra

Lemma 8.2 (Empirical law of a regularly varying spectrum)**.** *Let $a_m = m^\gamma L(m)$ be positive and regularly varying with index $\gamma > 0$, and put $M_n = \max_{1 \le j \le n} a_j$. Then*

$$\frac{M_n}{a_n} \longrightarrow 1$$

and

$$\frac{1}{n} \sum_{j=1}^{n} \delta_{a_j / M_n} \Rightarrow \nu_{1/\gamma}. \tag{73}$$

Proof. For every fixed $\delta \in (0,1)$, the uniform convergence theorem for regular variation gives

$$\sup_{x \in [\delta, 1]} \left| \frac{a_{\lfloor nx \rfloor}}{a_n} - x^\gamma \right| \to 0.$$

Potter bounds imply that for any $\epsilon \in (0, \gamma)$ there is C_ϵ such that, apart from finitely many indices,

$$\frac{a_j}{a_n} \le C_\epsilon (j/n)^{\gamma - \epsilon}, \qquad j \le n.$$

Since $M_n \ge a_n$, it remains to prove the matching upper bound. On $[\delta n, n]$ this follows from uniform convergence, while Potter bounds make $\sup_{j \le \delta n} a_j / a_n$ arbitrarily small after first choosing δ and then n large; the finitely many exceptional initial indices are $o(a_n)$. Hence $M_n / a_n \to 1$. For every bounded continuous φ, split the Riemann sum at δn. The small-index part has mass at most δ, while on $[\delta, 1]$ uniform convergence gives

$$\frac{1}{n} \sum_{j=1}^{n} \varphi(a_j / M_n) \longrightarrow \int_0^1 \varphi(x^\gamma)\, dx.$$

The latter is integration against $\nu_{1/\gamma}$. □

Proof of Corollary 1.7. The empirical normalized rate law is independent of the assignment to labels. Lemma 8.2 identifies its limit as $\nu_{1/\gamma}$, and Theorems 1.1 and 1.5 apply. □

For the exact power-law spectrum, the late-clock mass has the explicit decay

$$\overline{\nu}(U) = \alpha \int_0^1 r^{\alpha-1} e^{-rU}\, dr \asymp U^{-\alpha}. \tag{74}$$

Thus the uncontrolled-row term in orbit balance is $O(U^{-\alpha/2}) = O(U^{-1/(2\gamma)})$. This is qualitative here because the mesh and cutoffs are fixed before $n \to \infty$, but it identifies the correct endpoint leakage scale.

9 Spectral boundary and further questions

The spectral formulation separates three inputs: vanishing late-clock mass, local flatness within finite clock slabs, and a global mean-zero contraction. We first establish a broader sufficient condition for the channel gap and identify its slowly varying boundary. The necessity of a gap for Poisson–Dirichlet universality and the behavior of multiple rate scales remain separate questions.

Theorem 9.1 (Dyadic-doubling criterion)**.** *Let ν be a probability law on $(0,1]$, and put*

$$w_k = \nu((2^{-k-1}, 2^{-k}]).$$

Suppose that, for some constants $0 < a \le b < 1$ and every sufficiently large k, one has $w_k > 0$ and

$$a \le \frac{w_{k+1}}{w_k} \le b.$$

Then $q(\nu) < 1$.

Proof. Choose K so that the ratio bounds hold for $k \ge K$, and let $I_k = (2^{-k-1}, 2^{-k}]$. Put $x = 2^{-k}$ and $t \in [x^{-1}, 2x^{-1}]$. The contribution of the inner shells to $B_\nu(t)$ is at most

$$\sum_{m\ge 0} 2^{-k-m} w_{k+m} \le x w_k \sum_{m\ge 0} (b/2)^m.$$

For an outer shell I_{k-m} with $1 \le m \le k-K$, one has $r \le 2^m x$, $rt \ge 2^{m-1}$, and $w_{k-m} \le a^{-m} w_k$. Its contribution is at most

$$x w_k (2/a)^m e^{-2^{m-1}}.$$

The sum over $m \ge 1$ converges. The remaining core $(2^{-K}, 1]$ contributes at most $e^{-2^{-K}/x} = o(x w_k)$, since $w_k \ge w_K a^{k-K}$. Thus, uniformly on the stated time interval and for all sufficiently large k,

$$B_\nu(t) \le C x w_k. \tag{75}$$

If r, s lie in I_k, or in the adjacent shells I_k, I_{k-1}, then $rs \ge c_1 x^2$ and $r + s \le c_2 x$. Restricting (63) to $[x^{-1}, 2x^{-1}]$ and using (75) gives

$$R_\nu(r,s) \ge c/w_k.$$

After enlarging K if necessary, the core has positive mass. Restricting (63) to $t \in [0,1]$ gives a positive kernel lower bound joining the core to I_K and within the core. The within-shell and adjacent-shell estimates in (69)–(70) therefore hold in this setting. Lemma 7.2 and (71), with $\rho = b$, give

$$\mathrm{Var}_\nu(f) \le C_\nu \langle f, (I - C_\nu C_\nu^*) f\rangle.$$

Positivity of $C_\nu C_\nu^*$ now yields $q(\nu) < 1$. □

The criterion permits oscillating shell ratios and atomic or singular rate laws. It also implies Theorem 1.5, because a regularly varying lower tail of index $\alpha > 0$ has $w_{k+1}/w_k \to 2^{-\alpha} \in (0,1)$.

Theorem 9.2 (Slowly varying boundary)**.** *Let ν be a probability law on $(0,1]$. Suppose $F_\nu(x) = \nu((0,x])$ is positive for all sufficiently small x, tends to zero, and is slowly varying at zero. Then $q(\nu) = 1$.*

Proof. In the joint exponential channel, define

$$A_\delta = \{R \le \delta\}, \qquad B_\delta = \{T > 1/\delta\}, \qquad a_\delta = F_\nu(\delta).$$

The Laplace–Stieltjes Abelian theorem at index zero [2, Chapter 1] gives

$$\mathbb{P}(B_\delta) = \int e^{-r/\delta} \nu(\,dr) \sim a_\delta.$$

Also, $1 - e^{-z} \le z$ and Stieltjes integration by parts yield

$$\begin{aligned}\mathbb{P}(A_\delta \setminus B_\delta) &\le \delta^{-1} \int_{(0,\delta]} r\nu(\,dr) \\ &= F_\nu(\delta) - \delta^{-1} \int_0^\delta F_\nu(u)\,du = o(a_\delta).\end{aligned}$$

For the last equality, substitute $u = \delta v$ and apply dominated convergence to $F_\nu(\delta v)/F_\nu(\delta)$ on $0 < v < 1$: slow variation gives pointwise convergence to one, and monotonicity bounds the ratio by one. Consequently $\mathbb{P}(A_\delta \cap B_\delta) \sim a_\delta$, while both marginal probabilities are asymptotic to $a_\delta \to 0$. Hence

$$\operatorname{Corr}(\mathbf{1}_{A_\delta}, \mathbf{1}_{B_\delta}) \longrightarrow 1.$$

The channel norm $q(\nu)$ dominates each of these correlations and is at most one, proving the claim. □

Theorem 9.2 identifies a boundary of the strict spectral-gap hypothesis. It does not assert failure of the PD(1) conclusion in that regime; necessity of the gap remains the following question.

Question 9.3 (Necessity of the spectral gap)**.** *Is $q(\nu) < 1$ necessary for* PD(1) *universality uniformly over all deterministic arrangements with empirical rate law ν? If $q(\nu) = 1$, can one construct an arrangement whose macroscopic cycle partition is not* PD(1)*?*

The present proof genuinely loses regeneration when the mean-zero resolvent is unbounded. It is unknown whether that reflects a transition of the model or only a limitation of the method.

A useful continuum-of-scales test family is explicit. For $\beta > 0$, let

$$\nu_\beta(\,dr) = \frac{1}{\beta r} \mathbf{1}_{[e^{-\beta},1]}(r)\,dr. \tag{76}$$

This is the limiting normalized rate law of $\theta_{n,i} = \exp(\beta i/n)$.

Proposition 9.4 (Degeneration of the log-uniform channel)**.** *For every $\beta > 0$,*

$$q(\nu_\beta) \ge \frac{\beta}{\sqrt{\beta^2 + 2\pi^2}} = 1 - \frac{\pi^2}{\beta^2} + O(\beta^{-4}). \tag{77}$$

Proof. Write $R = e^{-X}$ with $X \sim \mathrm{Unif}(0, \beta)$. If $E \sim \mathrm{Exp}(1)$ is independent, then $T = E/R = Ee^X$, so $\log T = X + \log E$. Maximal correlation dominates the ordinary correlation of the centered functions $X = X(R)$ and $\log T$. Since $\mathrm{Var}\, X = \beta^2/12$ and $\mathrm{Var}(\log E) = \pi^2/6$,

$$q(\nu_\beta) \geq \mathrm{Corr}(X, \log T) = \sqrt{\frac{\mathrm{Var}\, X}{\mathrm{Var}\, X + \mathrm{Var}(\log E)}} = \frac{\beta}{\sqrt{\beta^2 + 2\pi^2}}.$$

□

Since ν_β is supported on $[e^{-\beta}, 1]$, the comparable-support bound gives $q(\nu_\beta) < 1$. Thus, for every $\beta > 0$, the norm gap satisfies

$$0 < 1 - q(\nu_\beta) \leq 1 - \frac{\beta}{\sqrt{\beta^2 + 2\pi^2}} \leq \frac{\pi^2}{\beta^2}.$$

The last inequality follows from $(1 + x)^{-1/2} \geq 1 - x/2$ for $x \geq 0$.

Question 9.5 (Growing log-linear spectra)**.** *For*

$$\theta_{n,i} = \exp(\beta_n i/n), \qquad \beta_n \longrightarrow \infty,$$

determine the largest-cycle scale and the maximal growth regime of β_n for which the PD(1) *limit persists. Is there a sharp transition between macroscopic and submacroscopic cycles?*

After normalization the empirical spectrum collapses to an atom at zero, so this problem is outside Theorem 1.1. Unlike a finite hierarchy of separated scales, it has a continuum of logarithmic rate scales. It is therefore a natural test of whether one-scale spectral universality can survive a slowly degenerating gap.

Conjecture 9.6 (Multiscale endpoint phase)**.** *Suppose that, after normalization by the largest rate scale, the empirical rate law has a positive atom at zero, and that the vanishing-rate sector admits one or more nontrivial limits after successive rescalings. Then the cycle partition has a hierarchical multiscale limit assembled from the Poisson–Dirichlet partitions of the successive rate scales, with a continuum-scale analogue when the logarithmic spectrum does not separate into finitely many levels.*

An atom at zero is excluded from Theorem 1.1 because a positive fraction of clocks escapes every fixed time cutoff. The conjecture asks for the correct replacement rather than a direct extension of the one-scale theorem.

Question 9.7 (Quantitative convergence)**.** *Under a uniform spectral gap and a polynomial bound $\overline{\nu}(U) \leq CU^{-\alpha}$, can one obtain an explicit rate of convergence to* GEM(1) *or* PD(1) *in Wasserstein or bounded-Lipschitz metric?*

Such a theorem would require growing partitions and quantitative approximation of the finite-state spectral gap. The present proof deliberately fixes all cutoffs and the mesh before sending $n \to \infty$.

Question 9.8 (Exact power-law gap)**.** *Determine the exact value, or sharp asymptotics as $\alpha \downarrow 0$ and $\alpha \to \infty$, of $q(\nu_\alpha)$. Is there a Mellin transform that diagonalizes an appropriate scale-invariant comparison channel?*

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